Mathur: We are focusing on tripling the renewable energy goals for 2030. So, if India is at, say, 140 or 150 GW as it was at the time of the G20 meeting, we are looking at approximately 500 GW now. I personally believe it will be much more than that.

TAPPING CLEAN SOURCES

Mathur: For vast areas of the world without electricity, solar mini grids are important. Because of a fall in prices in solar PV and batteries, we can look at a trade-off. So, if you have to extend the grid by more than 10 km, solar plus battery is a cheaper option. And, therefore, the only way that we can meet the provision of electricity to all by 2030 is by solar mini grids.... We also have to focus on financing. In 2022, around $310 billion (Rs 25.9 lakh crore) worldwide was invested in solar power... this year, it'll be about $380 billion (Rs 31.7 lakh crore), but hardly 3 per cent goes to Africa. India has a huge learning. We created a payment guarantee mechanism in the Solar Energy Corporation of India, which has enabled a lot of solar capacity to be developed. Investors came in, who saw this as an opportunity in which they would get their returns on time. We need a global payment guarantee mechanism, especially for Africa to pull in money there.

> "The only way we can provide electricity to all by 2030 is by solar mini grids. We also have to focus on financing it, as it requires a lot of investments. In 2022, around $310 billion (Rs 25.9 lakh crore) worldwide was invested in solar power, which compares favourably with peak investment in electricity generation"
>
> **AJAY MATHUR**
> Director General,
> International Solar Alliance

Bhalla: The government announced the National Green Hydrogen Mission in January this year, and came out with the bid for manufacturing electrolysers [apparatus that splits water into hydrogen and oxygen molecules using electricity] as well as green hydrogen production. Its budget is around Rs 20,000 crore till 2030. The government is hopeful that by mid-2024, Rs 17,500 crore of subsidies would be allocated to companies.... Other than the incentive, the Centre's R&D roadmap will also help in this. Moreover, it is assumed that about 60-70 per cent of India's green hydrogen would be exported.

THE STORAGE STORY

Bhalla: Battery is going to be an increasing mainstay of renewable energy. Our estimate is that by 2027, we would start needing about 50 GWh of installed battery capacity, which would rise to about 250 GWh by 2030. So, we have to step on it.

Mathur: Storage is not an electricity generation source of its own. It provides reliability.... There are three ways in which the lowering of costs occurs. First, new deposits are found. We are finding more lithium where we did not expect it, like in Jammu & Kashmir and Rajasthan. Secondly, because of competitive pressures, the amount of lithium per battery reduces. Thirdly, new kinds of batteries come up. So instead of lithium, you have sodium ion batteries. Commercialisation will depend on scale. ■



RENEWABLE ENERGY
A PARADIGM SHIFT

CLEAN POWER IMPERATIVE

India's green energy transition demands a sweeping overhaul of the sector—from infrastructure to financing and from hybrid systems to favourable regulations